

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

CON316 - R

Amazon ECS auto scaling deep dive

Ratnopam Chakrabarti

(he/him)

Containers Specialist Solutions Architect
AWS

Sushanth Mangalore

(he/him)

Sr. Solutions Architect
AWS



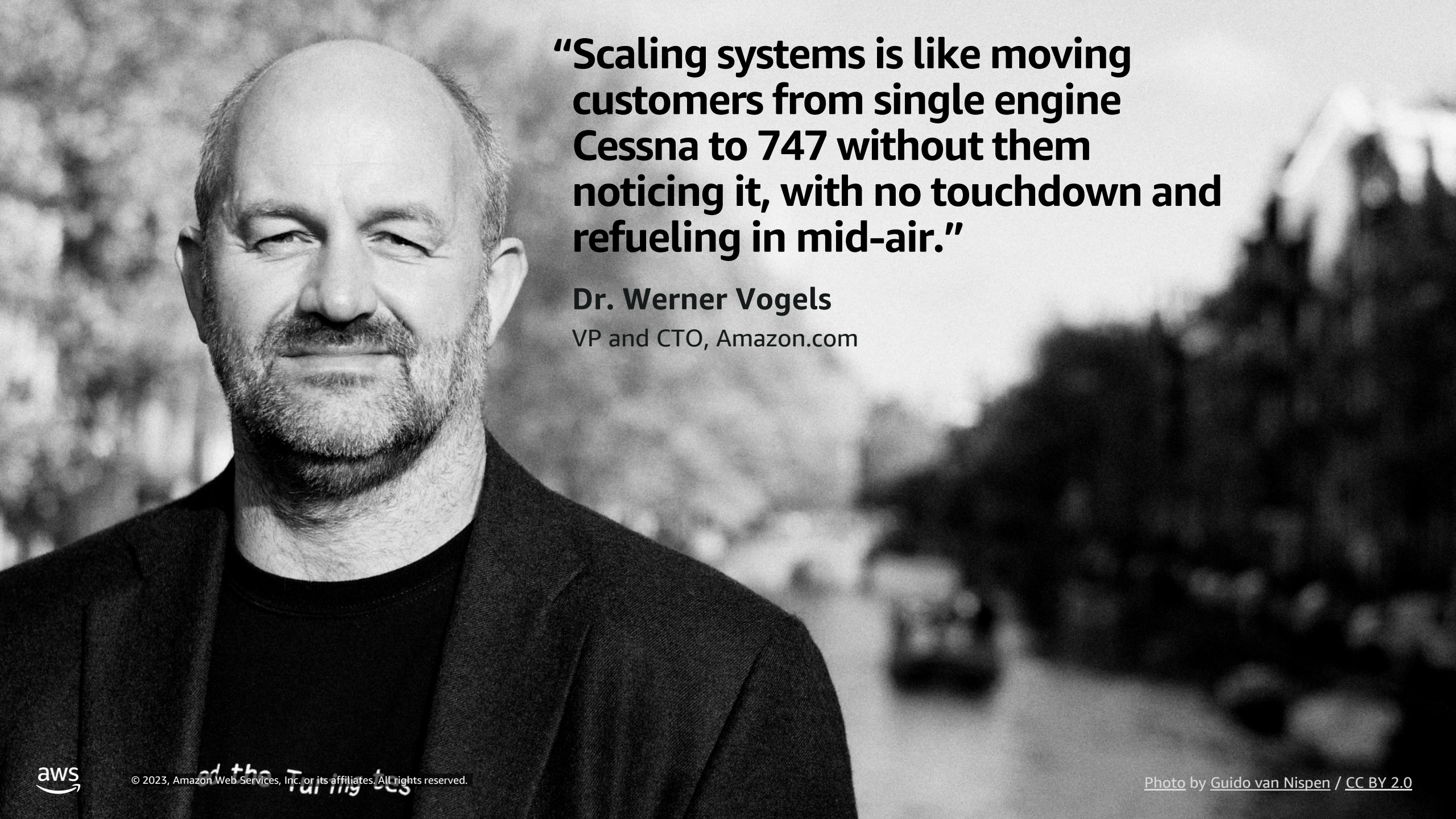
© 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agenda

- 01 Significant scale of Amazon ECS
- 02 Why scaling matters
- 03 Scaling your applications on Amazon ECS
- 04 Cluster capacity scaling
- 05 Best practices for efficient scaling
- 06 Q&A and whiteboarding
- 07 Call to action

Please ask us questions



A black and white portrait of Dr. Werner Vogels, a middle-aged man with a beard and mustache, wearing a dark jacket over a dark t-shirt. He is looking slightly to the right with a slight smile. The background is a blurred outdoor scene with trees and a building.

“Scaling systems is like moving customers from single engine Cessna to 747 without them noticing it, with no touchdown and refueling in mid-air.”

Dr. Werner Vogels

VP and CTO, Amazon.com



Amazon ECS: Significant scale



Over 2.25 billion

tasks launched every week

Tens of thousands

of API requests served per second

32 AWS Regions, 6 continents

Global presence

**65% of all new AWS
containers customers**

use Amazon ECS

Amazon and AWS services innovate with Amazon ECS



Amazon Redshift



Amazon OpenSearch Service



Amazon RDS



Amazon Transcribe



AWS Elemental MediaLive



Amazon.com recommendations



Amazon Lex



Amazon Polly



AWS Batch



Amazon SageMaker



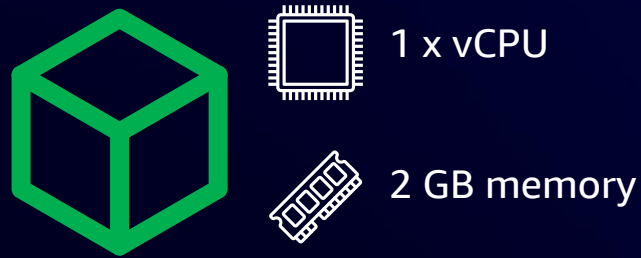
Amazon Translate

Amazon ECS delivers security, reliability, availability, and **scale**

Scaling your applications on Amazon ECS



With container orchestration, you focus on the application and **treat compute as generic capacity**



Here is my application container, and I want to launch it as a service with 4 copies



Amazon ECS

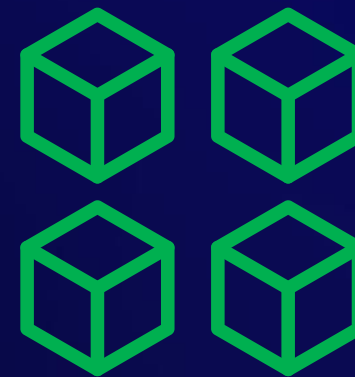


Here is a collection of Amazon EC2 instances to use as capacity

AWS Fargate lets you run containers directly: No Amazon EC2

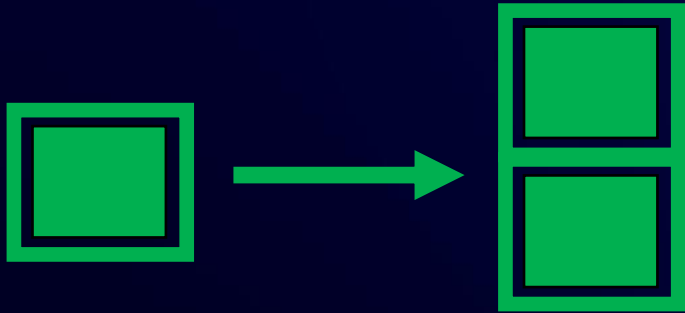


Here is my application
container, and I want
to launch it as a service
with 4 copies

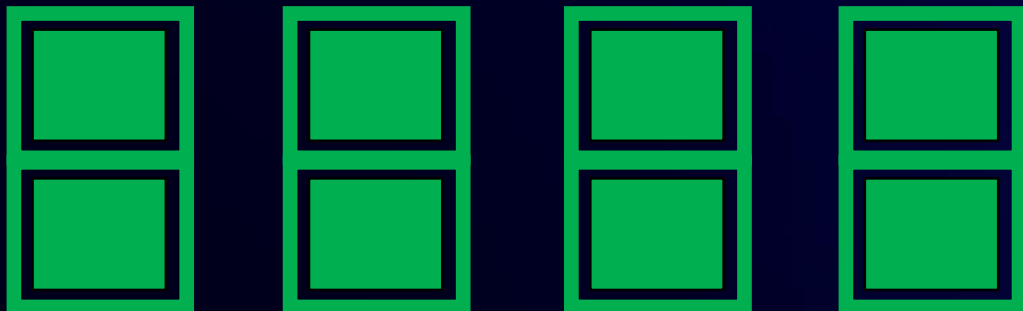


Amazon ECS scaling options

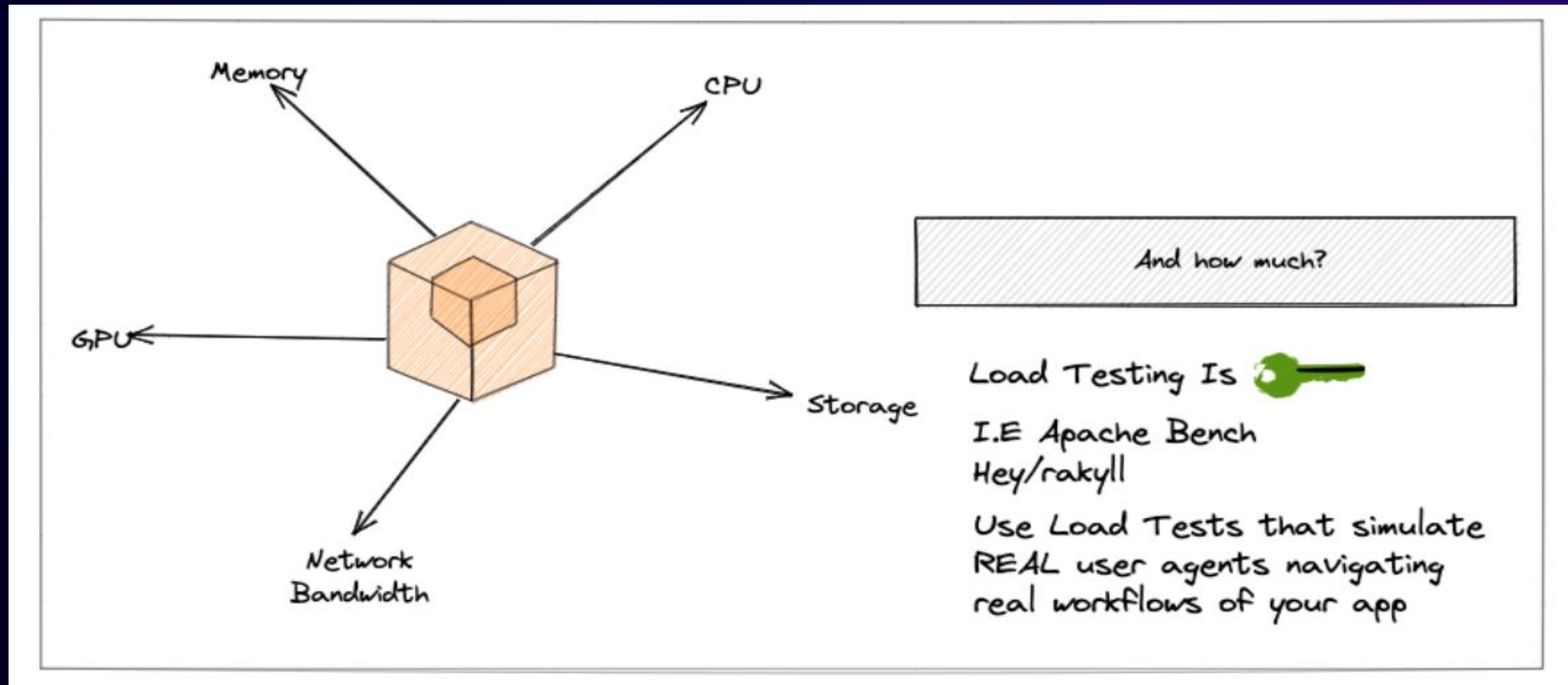
Vertical scaling: Increase size of the application container



Horizontal scaling: Run more copies of the application container



Vertical scaling



Amazon ECS integrates with AWS Application Auto Scaling to automatically scale your service horizontally

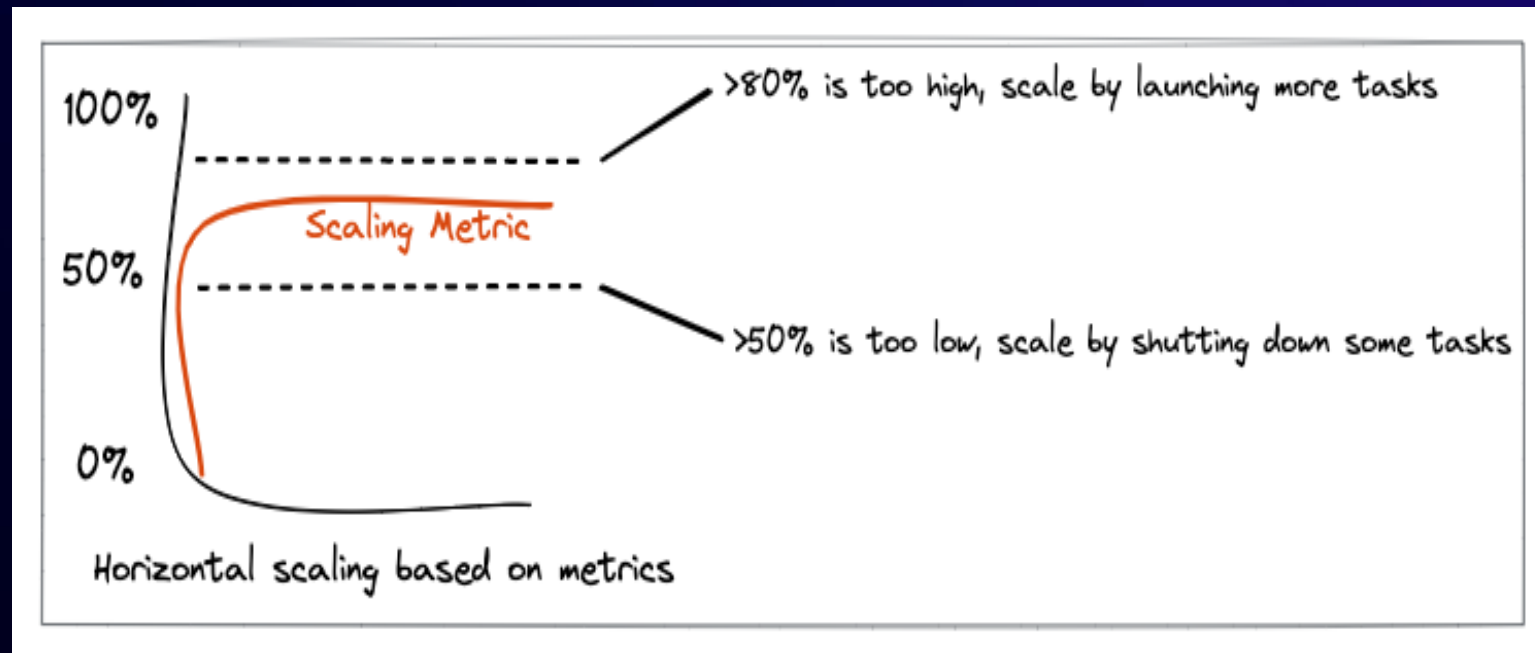


Aggregate metrics
from the service

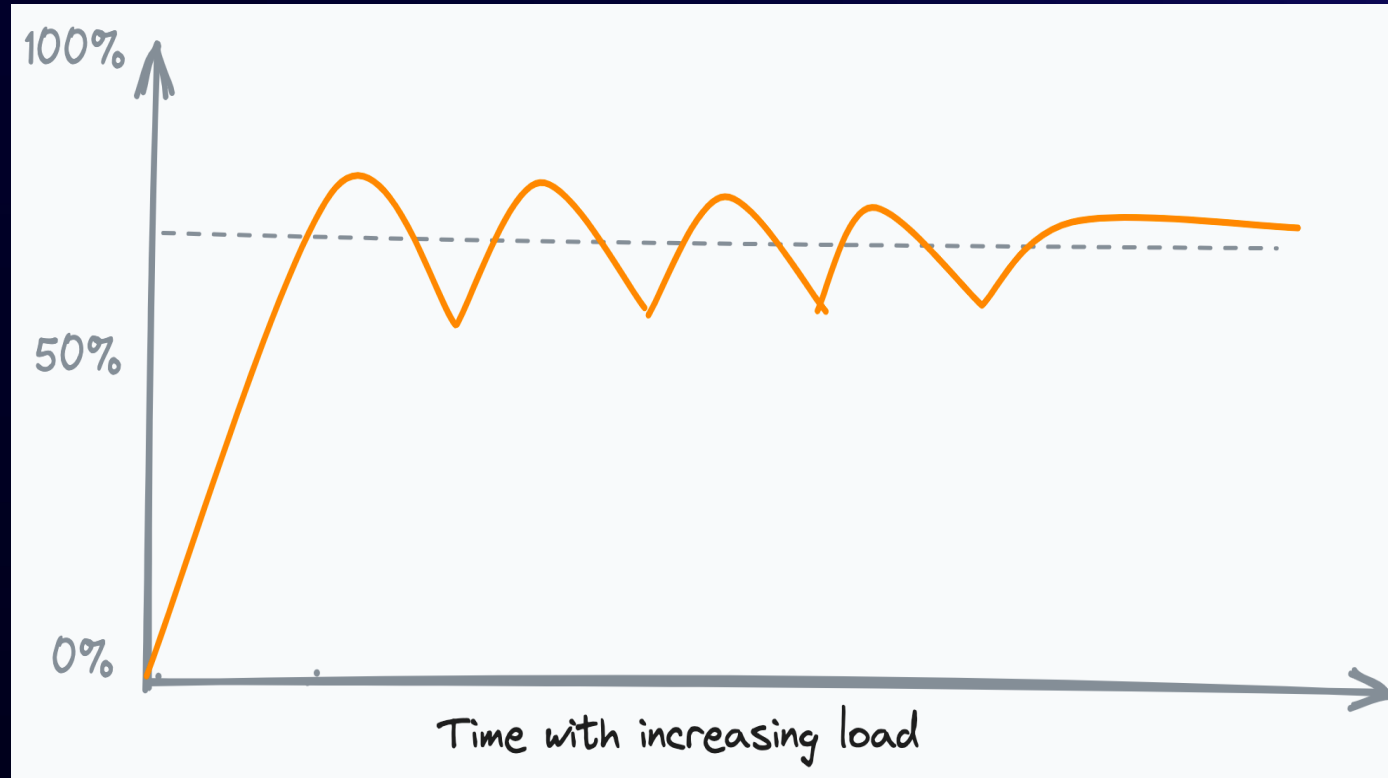
Auto scaling
policy is triggered

Amazon ECS launches
more tasks or stops
tasks according to the
number of tasks that
the policy wants

Horizontal scaling is typically based on aggregate resource consumption metrics for the service



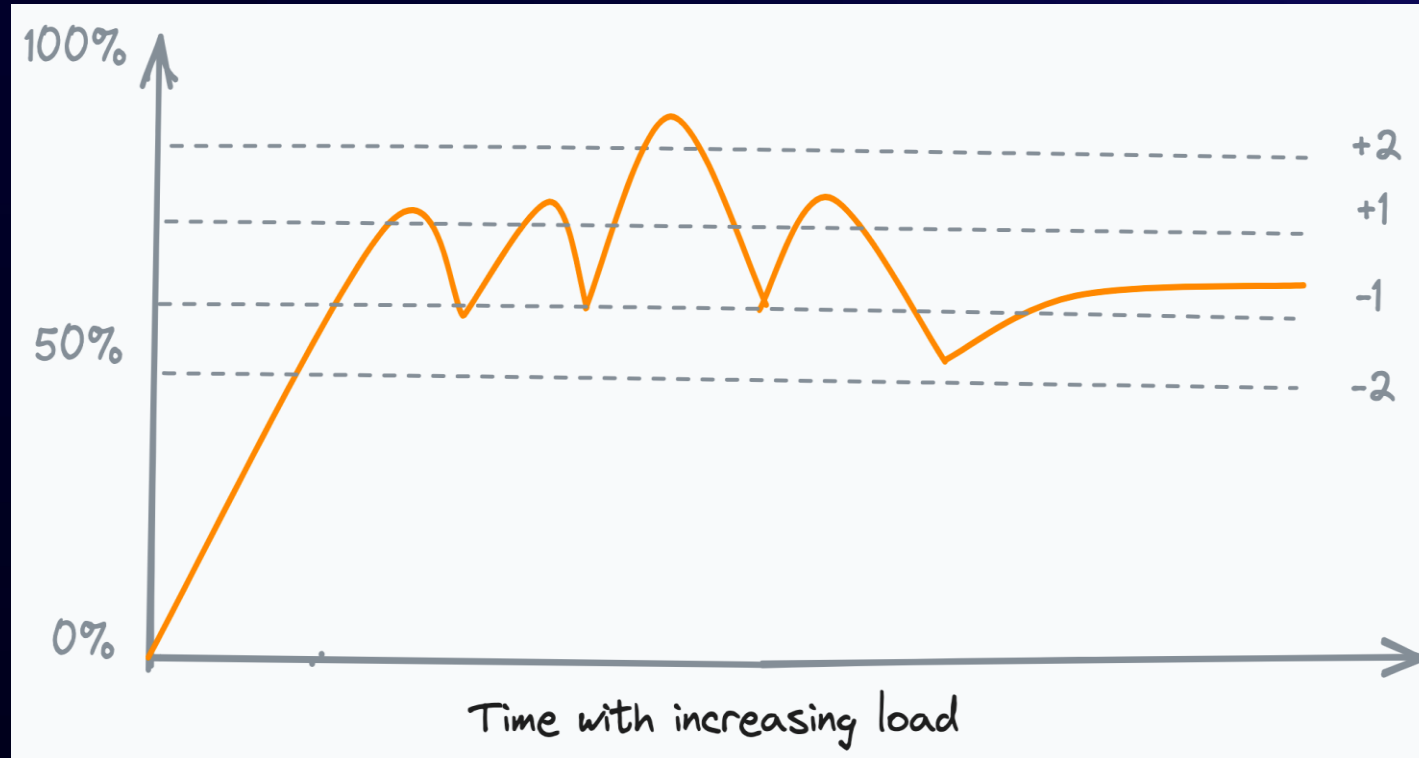
Types of scaling



Target tracking scaling policy:

Target tracking – One metric threshold, scale proportionally

Types of application scaling policies available

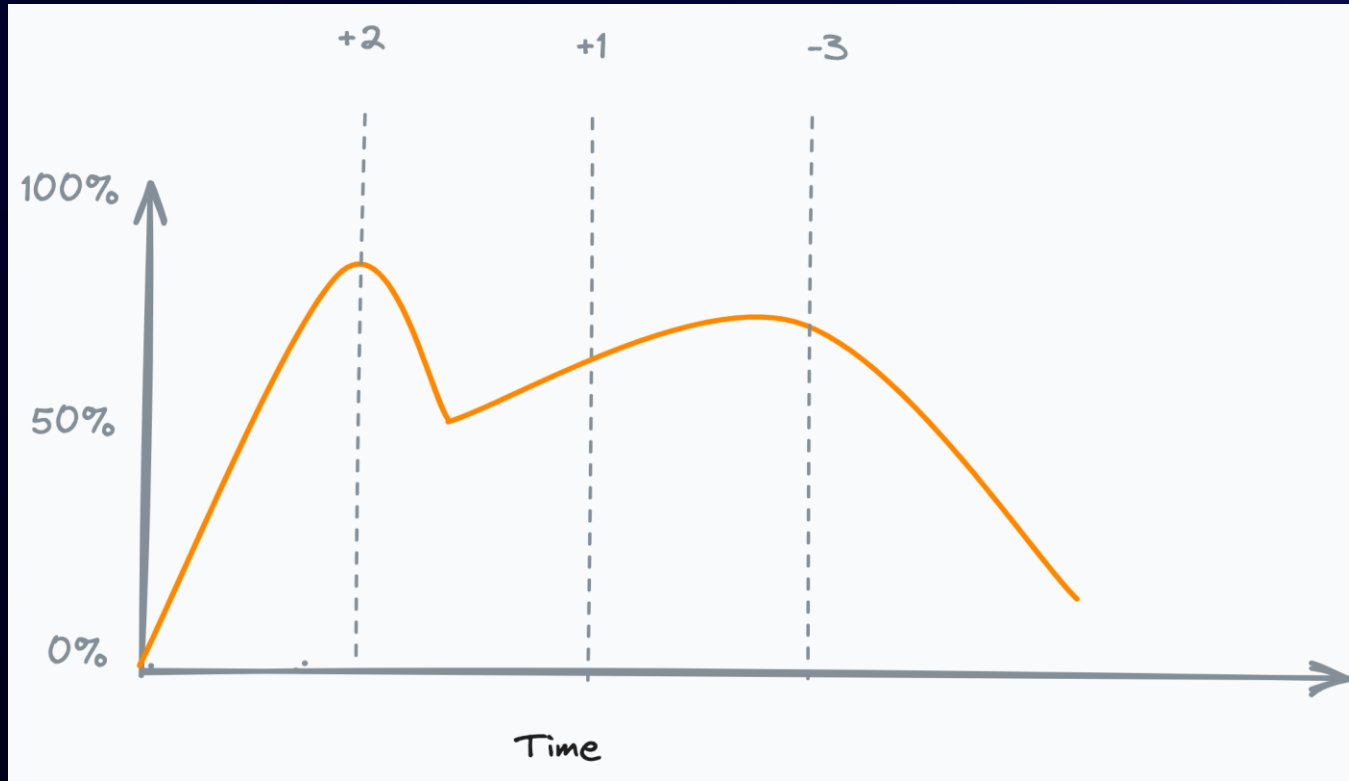


Step scaling policy:

Multiple bounds

Custom values for how many tasks to add or remove

Types of application scaling policies available

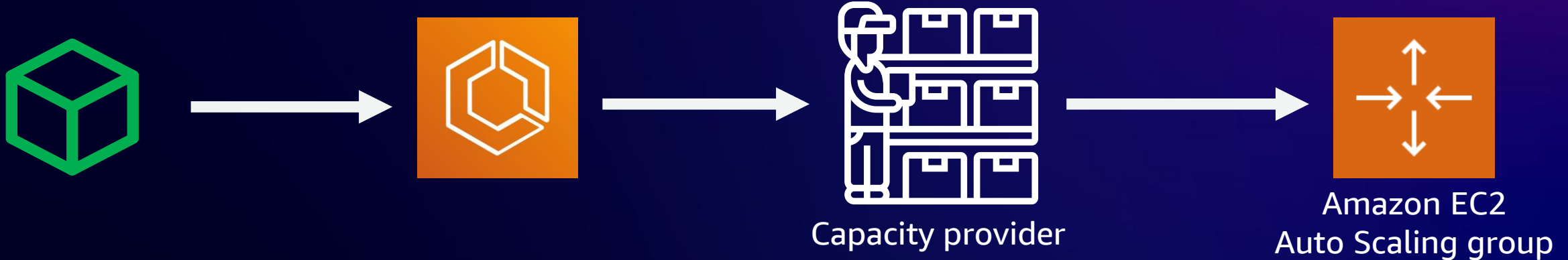


Scheduled scaling policy:

Used for services with predictable, static pattern of daily traffic

Cluster capacity scaling

Capacity providers



"Run 10 copies of my container"

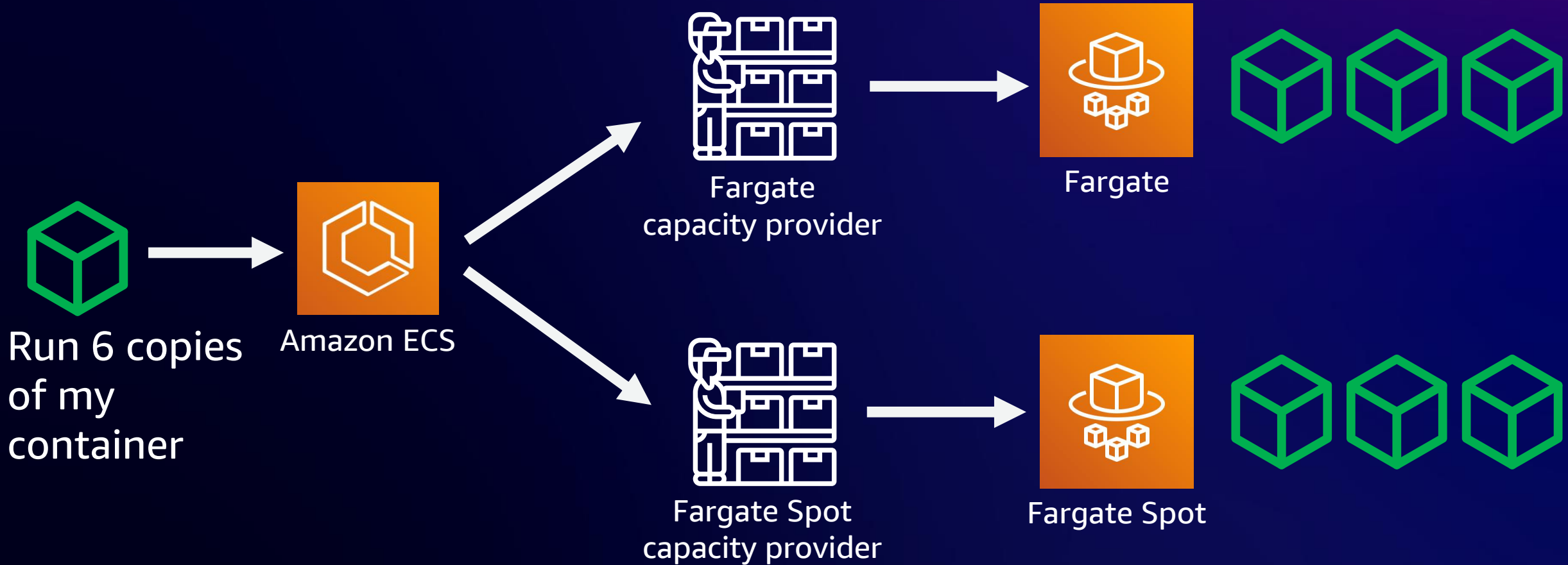
"Okay I'll launch those 10 containers"

"I see that cluster reservation is exceeding limits, so we need two more instances"

"I will launch two more Amazon EC2 instances"

Capacity providers

YOU CAN HAVE MULTIPLE CAPACITY PROVIDERS



Best practices for efficient scaling

Good and bad scaling metrics

✓ Average CPU utilization

Many (most?) applications are CPU bound

✓ Average queue depth

More workers lead to faster work consumption

? Average response time

Adding more tasks doesn't always help;
in fact, it can make a problem worse

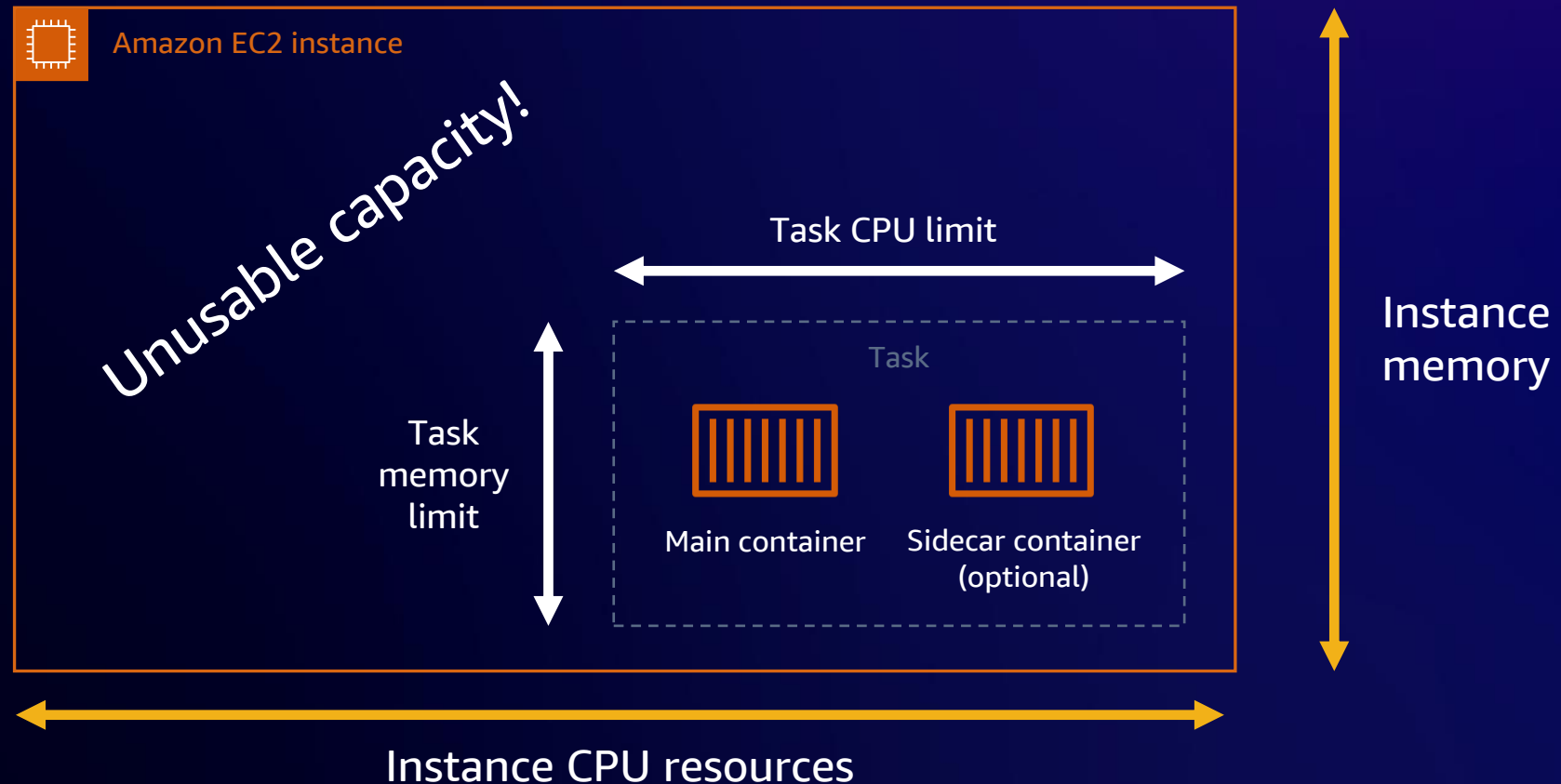
✗ Memory utilization

✗ max(N)

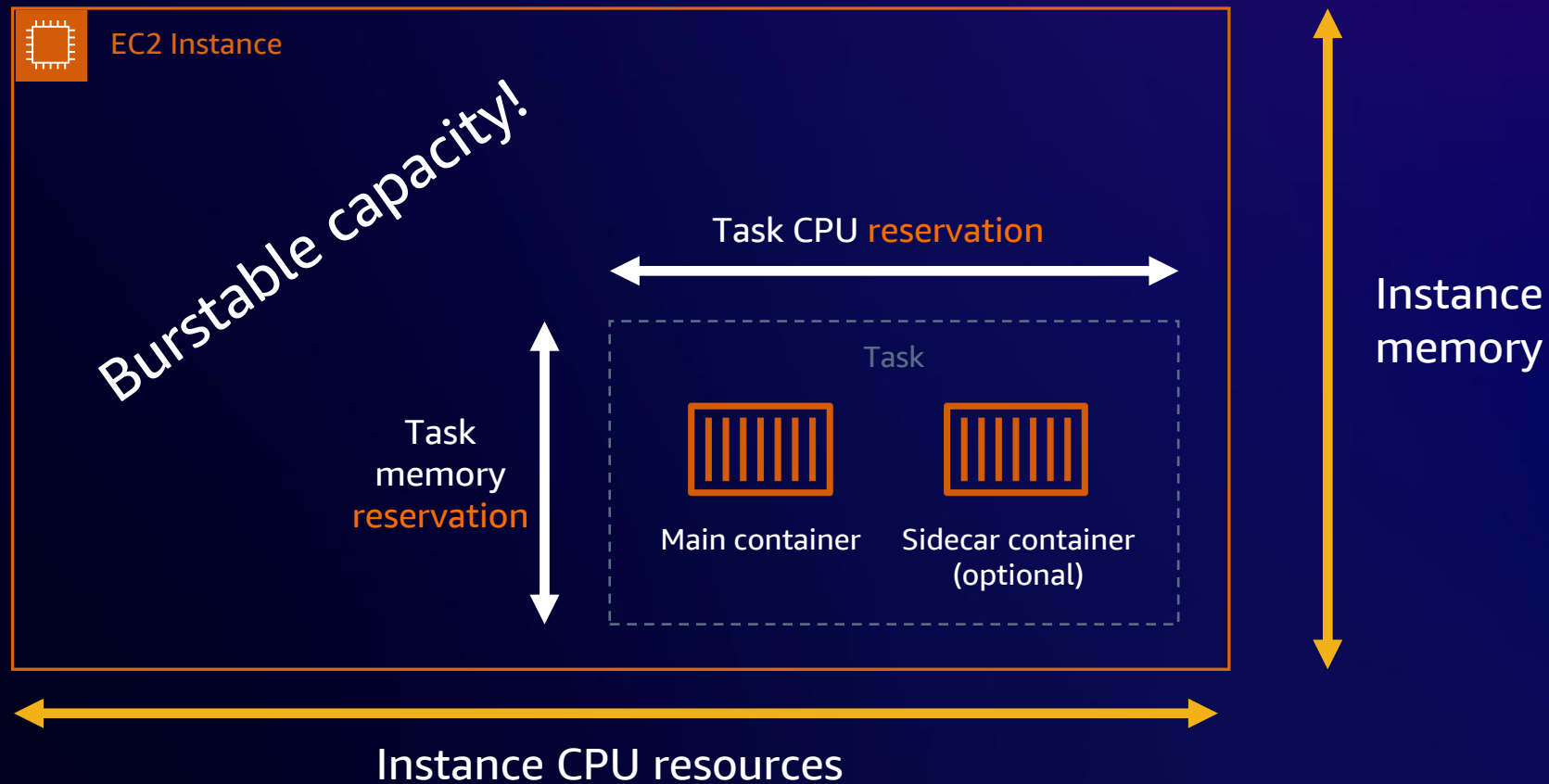
✗ p-values (for example, p95, p99)

Resource limits

Task sizes on Amazon EC2: **Not recommended**



Amazon EC2: Use container **reservations** instead of task limits



Q&A



Call to action



**Amazon ECS
scaling diagrams**



**Amazon ECS
workshop**



**Amazon ECS
best practices**

Thank you!



Please complete the session survey in the mobile app

Sushanth Mangalore
mangalos@amazon.com

 sushanthmangalore

Ratnopam Chakrabarti
aratnch@amazon.com

 ratnopamc